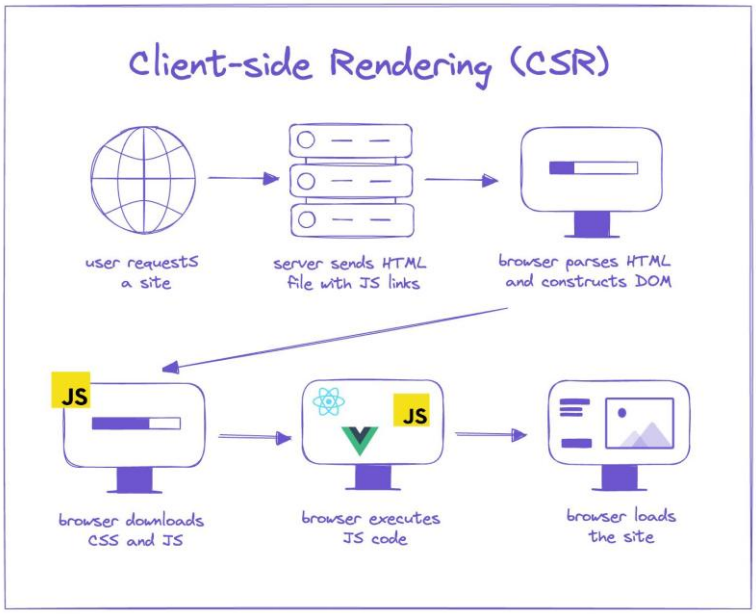
JavaScript (JS)

JavaScript (JS) is a **programming language used to make websites interactive and dynamic**. It runs mainly in the **web browser** and allows web pages to respond to user actions.

Key Features of JavaScript

1. **Client-side language** – Runs in the browser.
2. **Event-driven** – Reacts to user actions (click, scroll, type).
3. **Dynamic typing** – Variable types can change.
4. **Used for full-stack development** with technologies like:
    - Node.js (backend)
    - React / Angular / Vue (frontend)

Client-Side Rendering

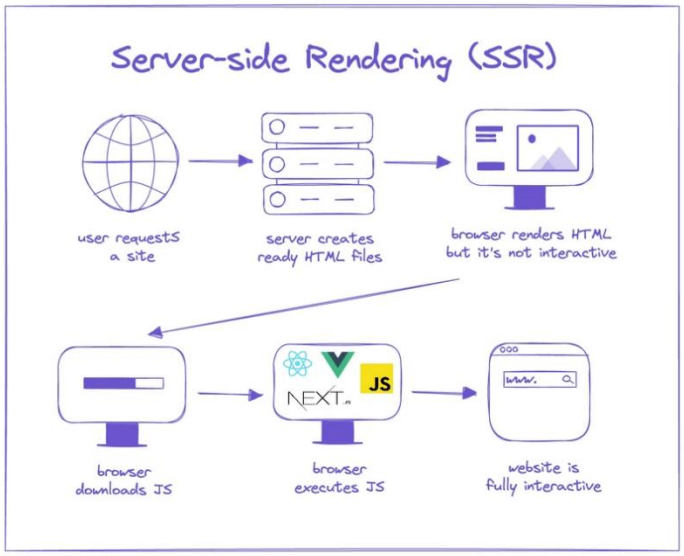


Simple Steps

1. **User opens a website**
   Browser sends request to server.
2. **Server sends basic HTML + JavaScript files**
3. **Browser downloads JavaScript**
4. **JavaScript runs in the browser**
5. **JavaScript creates the webpage content**

6. **User sees the website**

Server-Side Rendering (SSR)



Simple Steps

1. **User opens a website**
   Browser sends request to the server.
2. **Server builds the complete HTML page**
3. **Server sends the ready webpage to the browser**
4. **Browser shows the page immediately**
5. **JavaScript loads later to make it interactive**

Frontend Frameworks

Frontend frameworks are used to **build the user interface (UI)** of a website.

Examples:

1. **React**
2. **Angular**
3. **Vue.js**
4. **Next.js**

Backend Frameworks

Backend frameworks are used to **handle server logic, databases, and APIs**.

Examples:

1. **Node.js**
2. **Express.js**

Second Version of Backend:

1. **Django**
2. **Flask**
3. **FAST & REST APIs**

<script> Tag

The **<script> tag** in **HTML** is used to **add JavaScript code to a webpage**.

Basic Syntax

```
<script>
  JavaScript code
</script>
```

Example:

```
<script>
  console.log("Hello Sudesh");
</script>
```

Variables in JavaScript

A **variable** is a **container used to store data** that can be used later in a program.

Example:

let name = “Sudesh”;

let age = 22;

Data Types in JavaScript:

A **data type** defines the **type of value stored in a variable**.

JavaScript data types are divided into **two categories**:

1. **Primitive Data Types**
2. **Non-Primitive Data Types**

1. Primitive Data Types

Primitive types store **single simple values**.

Types of primitive data types:

1. Number

Used for numbers.

```
let age = 22;
let price = 99.5;
```

2. String

Used for text.

```
let name = "Sudesh";
```

3. Boolean

Represents **true or false**.

```
let isLoggedIn = true;
```

4. Undefined

A variable declared but **not assigned a value**.

```
let x;
```

Output:

```
undefined
```

5. Null

Represents **empty or no value**.

```
let data = null;
```

6 Symbol

Used for **unique identifiers**.

```
let id = Symbol("id");
```

2. Non-Primitive Data Types

These store **multiple values or complex data**.

1. Object

```
let student = {
  name: "Sudesh",
  age: 22
};
```

2. Array

Used to store **multiple values in a list**.

```
let numbers = [1,2,3,4,5];
```

3. Function

A block of reusable code.

```
function greet() {
  console.log("Hello");
}
```